Varun Ganjigunte Prakash

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PROFILE

Computer vision and applied AI engineer with production experience in multimodal video understanding, object detection fine-tuning, document AI, and speech biometrics for clinical healthcare. Five peer-reviewed publications in computer vision for autism assessment and human action recognition.

EDUCATION

Bachelor of Engineering in Electronics and Communication Aug 2014 - May 2018
Sri Jayachamarajendra College of Engineering *Mysuru, India*
GPA: 8.47/10.0

WORK EXPERIENCE

Senior Research Scientist Jan 2025 - Oct 2026
Frontera Health *Remote / India*

Senior Computer Vision Engineer *Dec 2023 - Feb 2025*

- Built and productionized multimodal document-AI pipelines using Gemini vision, PDF rasterization, typed schemas, and gold-set regression tests, achieving 94–98% field accuracy across clinical assessments and expanding support to ABLLS, AFLS, ADI-R, CARS-2, Intake, and RIAS-2.
- Fine-tuned RF-DETR on COCO-formatted assessment grids and integrated it into an ABLLS vision parser, reaching 0.772 AP and 0.936 AP50 on an unseen test split versus 0.583 AP and 0.846 AP50 for the strongest SSD baseline.
- Designed an end-to-end speaker-signature pipeline using Silero VAD, pyannote diarization, WeSpeaker embeddings, similarity-graph clustering, and human review, enabling learner/therapist voice registration and achieving 80.2% segment accuracy and 92.2% mean person-level accuracy in clinic evaluations.
- Productionized a multimodal scream/cry detection pipeline combining HumanOmni video-audio inference, YAMNet filtering, and Gemini validation; evaluated 59.1 hours across 65 therapy videos and contributed to a documented test precision/F1 of 0.785/0.751.
- Led evaluation of multimodal and pose-based approaches for rare maladaptive behaviors in long therapy videos, comparing VideoMamba, YAMNet, HumanOmni, and RGB+pose pipelines; identified a candidate cascade that removed 91.5% of normal chunks while retaining 73% of positives.

Additional contributions

- Redesigned multimodal PDF parsing into a Temporal fan-out/fan-in architecture with page-level retries and durable checkpoints, reducing five-page processing time by 1.92–2.15× while preserving structured output compatibility.
- Built a child/adult detection fine-tuning pipeline over therapy and open-source data, benchmarking production ONNX against RF-DETR, YOLOv8, and YOLO-World v2; the best fine-tuned model reached 0.395 AP50 and 74.2% child recall versus 0.245 AP50 and 51.7% for the production baseline on a held-out test split.
- Improved social-gaze estimation in a therapy-video engagement prototype by fusing gaze-model outputs with YOLOv4 head detection and WHENet head-pose estimation, using rule-based correction to reduce false child–therapist look-at predictions.
- Implemented clinical session-summary aggregation in the social-engagement stack, combining chunked video-scene Gemini prompts, structured outputs, and multi-stage text consolidation for learner-level therapy session narratives.

- Built an end-to-end maladaptive-behavior annotation system selecting high-signal therapy sessions, chunking video via AWS Batch, and delivering a review tool with chunk manifests, persistent storage sync, and clinician-facing labeling workflows.
- Productionized environment-driven routing and credential handling for fine-tuned Gemini vision models used in InternVideo2-backed aggressive-behavior classification, enabling region-aware deployment of maladaptive VLM inference.
- Developed an embedding- and LLM-based clinical text style-transfer toolkit with PII anonymization, author-profile clustering, and evaluation visualizations to support report-writing and prompt-style experiments.
- Added gold-set regression tests and a field-level confidence review layer for multimodal assessment parsers, improving release safety for clinical document-AI pipelines across ABLLS/AFLS and shared image-parsing harnesses.
- Extended the clinical report-builder pipeline to inject provider language preferences into Gemini prompt inputs via FDM GraphQL, improving localization-aware assessment report generation.
- Prototyped a zero-shot V-JEPA 2 hit/throw behavior baseline on therapy-video clips, with reproducible smoke manifests, probability aggregation, and HTML review tooling for rare action-recognition experiments.

- Built evaluation and DOCX generation tooling for BCBA supervision-note LLM outputs, including structured scoring and report export for clinical documentation experiments.
- Contributed to the production social-engagement computer-vision stack spanning person tracking, child/adult detection, gaze models, and Gemini-based engagement summarization, including early pipeline integration, visualization tooling, and cross-model evaluation on clinic video.
- Hardened GPU-intensive social-engagement and behavior workflows by adding explicit tensor cleanup, Temporal heartbeats, failure fallbacks, and environment-driven model versioning, improving reliability across person, speaker, and video-classification activities.
- Improved clinical-video ASR by resolving ghost transcripts through speech-segment language voting and later rebuilding subtitles from diarized utterances with word-level timing, overlap handling, and readability limits.

- Built and iterated a human-in-the-loop validation platform for TACT and verbal-operant models, combining synchronized video/transcript review, S3-backed audit data, target-coverage analysis, and clinician feedback workflows.
- Developed a Gemini- and spaCy-based vocabulary monitoring pipeline over speaker-tagged transcripts, producing clinician-facing measures such as type-token ratio, mean length of utterance, and syntactic diversity.
- Developed a clinical goal-recommendation research stack spanning popularity and ALS baselines, neural collaborative filtering, sequence models, and NER over VB-MAPP taxonomies with learner-disjoint evaluation.

Senior Engineer, Data Science Mar 2023 - Dec 2023
Colt Technology Services *Bengaluru, India*

- Implemented and integrated NLP models for Contract Data Mining.
- Implemented and integrated NLP models for Sentiment Analysis.
- Supported and collaborated with the development of ML models for Opportunity Prioritization.
- Conducted experiments and evaluated the feasibility of implementing cutting-edge NLP techniques that ensure scalability, provide the best user experience, and foster wide adoption and innovation for various stakeholders.
- Contributed to biweekly ML and NLP technical sessions to foster collaboration and learning within the team.

Head Machine Learning and Technology Architect July 2022 - Feb 2023
CogniAble *Remote, India*

- Researched and implemented a few computer vision approaches for clinical screening.

- Implemented a neural network-based recommender system for treatment personalization and skill enhancement for patients with neurodevelopmental disorders

Machine Learning Engineer - Computer Vision Mar 2021 - July 2022
CogniAble *Remote, India*

- Trained and deployed a custom action recognition deep network for recognizing children's behaviors.
- Implemented ML classifiers for neurodevelopmental disorder patients' treatment personalization and skill enhancement (classical, recommender systems, and graph models).
- Developed object classifiers and trackers for children's skill recognition in clinical settings.
- Led and managed the development of computer vision models in production (9 models).
- Implemented large-scale data processing, model training, and inference pipelines. Managed a team of data engineers and clinicians.
- Researched and implemented several novel computer vision approaches for clinical screening.

Engineer June 2018 - Mar 2021
L&T Technology Services *Bengaluru, India*

- Developed three computer vision-based software applications for edge inference products for a leading customer in AI and semiconductors.
- Proposed and implemented improvements to existing computer vision frameworks.
- Software development for a medical sterilization product in C++ and Lua.
- Explored facets of AI subsets and proposed solutions (POC) to establish strong customer relationships from ideation to realization.
- Demonstrated the applicability of AI and initiated efforts to implement AI in the semiconductor industry.
- Designed and implemented an OCR pipeline for HMI screen automation (2x time savings).
- Developed and evaluated performance enhancements for customer's deep models using the OpenVINO.
- Taught engineers about deep learning for computer vision and its applications.
- Collaborated to assist teams in resolving software problems and simplifying them through automation.
- Implemented an auto-scripting code that generates codes (Software 2.0), which reduced manual tasks from hours to minutes (10x faster). Some code implementations were effective, and the unsuccessful ones taught me valuable lessons.
- Stood out among the few hundreds of software development and testing teams in the transportation business unit as the only one willing to automate and accelerate software processes using open-source tools.
- Improved several business processes with computer vision-based automation. Automated most simple, repetitive tasks yet time-consuming in project development and delivery.

Computer Vision Intern Feb 2018 - May 2018
Skylark Drones *Bengaluru, India*

- Researched and developed tools for an aerial image overlap checker to address drone terrain variation issues.

Embedded Systems And Wireless Network Intern June 2016 - Dec 2016
LogicHive Solutions *Mysuru, India*

- Worked on various projects that involved multifarious sensors to implement practical applications with Ethernet, Bluetooth, WiFi, and other networking principles.
- Developed a few real-world wireless applications for GPS geo-fencing with ZigBee, robot control, RF communication, and an electronic liquid measuring scale.

PROJECTS

Behavior-aware robot navigation with deep reinforcement learning

- Implemented a human behavior-aware crowd navigation model utilizing the DRL algorithm with enhanced risk awareness, safety, and understanding of the environment for robot navigational decisions.

6-DOF pose estimation, planning, and control

- Designed a custom 4–axis robotic arm, implemented it, and simulated it in ROS.
- Tested the custom-built manipulator for 3D object grasp with deep learning.

Indian currency recognition and food classifier application

- Implemented a deep model for currency recognition and deployed it as an Android mobile application.
- Developed a deep model that classifies and discovers related information about the 20 most common Indian food items, such as the nearest restaurants where they are available and the item's cost.

Autonomous mobile manipulator robot: Dexterous Service Robot *Advances in Robotics, 2019*

- Designed a home assistant robot to assist differently-abled and elderly persons by performing some of the common tasks involved in daily life.
- Implemented the objectives using a robot (DEXTER ER-2) mounted on a mobile vehicle that operated using voice commands to selectively search for and deliver the object required by the user.

Robot to sort objects

- Implemented code to sort different-sized objects based on color with a robotic arm (DEXTER ER-2). The algorithm used basic image processing and inverse kinematics concepts.

Robot to pick and deliver objects

- Designed and implemented object delivery behavior for a robot based on the shape, size, and color of the objects.
- Implemented perception algorithms to analyze objects and obstacles and plan paths.
- Evaluated objectives using the Fire Bird V robot with a mounted custom-designed gripper to pick and deliver objects.

Miscellaneous

- *Biomorphic Hyper–redundant Snake Robot* (Oct 2017 – Jan 2018): Designed and built a robot resembling a snake in Autodesk Fusion 360. Simulated different gaits such as serpentine, caterpillar, and side-winding motion in V–REP (now CoppeliaSim). The project was a part of the e-Yantra robotics competition in 2018.
- Implemented transmitter and receiver codes with an encrypted communication system (Morse code) for information communication and validated them on two Zigbee modules.

PUBLICATIONS

- "Computer Vision-Based Assessment of Autistic Children: Analyzing Interactions, Emotions, Human Pose, and Life Skills", **Varun Ganjigunte Prakash**, Manu Kohli, Swati Kohli, Prathosh A. Prasad, Tanu Wadhera, Diptanshu Das, Debasis Panigrahi, John Vijay Sagar Kommu, *IEEE Access, 2023*

- "Deep Learning-Based Human Action Recognition Framework to Assess Children on the Risk of Autism or Developmental Delays", Manu Kohli, Arpan Kumar Kar, **Varun Ganjigunte Prakash**, Prathosh A. Prasad, *International Conference on Neural Information Processing (ICONIP), Communications in Computer and Information Science (CCIS), Springer, 2023*

- "Video-based real-time assessment and diagnosis of autism spectrum disorder using deep neural networks", **Varun Ganjigunte Prakash**, Manu Kohli, Prathosh A. Prasad, Monica Juneja, Manushree Gupta, Smitha Sairam, Sadasivan Sitaraman, Anjali Sanjeev Bangalore, John Vijay Sagar Kommu, Lokesh Saini, Prashant Ramesh Utage and Nishant Goyal, *Expert Systems, Wiley Online Library, 2023*

- "Behavior-Aware Robot Navigation with Deep Reinforcement Learning", **Varun Ganjigunte Prakash**, *IEEE 6th International Conference on Computation System and Information Technology for Sustainable Solutions (CSITSS), 2022*

- "Autonomous Service Robot", Arshad Javeed, **Varun Ganjigunte Prakash**, Sudarshan Patilkulkarni, *Advances in Robotics (AIR 2019) – ACM ICPS, 2019*

TECHNICAL SKILLS

Languages:	Python, C++, MATLAB, C, Embedded C
Computer Vision:	Object detection (RF-DETR, YOLO-World, YOLOv8, SSD), action recognition, pose and head-pose estimation, person tracking, OpenCV, COCO evaluation
Multimodal & GenAI:	Gemini Vision, HumanOmni, InternVideo2, VideoMamba, V-JEPA, YAMNet, prompt engineering, model ensembling
Speech & Audio:	WeSpeaker, Wav2Vec2, pyannote, Silero VAD, speaker diarization, ASR, FFmpeg
ML Frameworks:	PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, ROS
MLOps & Data:	Temporal, AWS (S3, Batch), Docker, CUDA, Git, Weights & Biases, Label Studio, V7 Darwin, gold-set regression testing
Interfaces:	Jetson Nano, Raspberry Pi, custom mobile robots and robotic arms

ACCOLADES AND LEADERSHIP

- Completed *DELF B1* (Advanced) certification for a Diploma in French Language administered by the International Centre for French Studies for France's Ministry of Education.
- Reviewer for Biomedical Signal Processing and Control and International Journal of Machine Learning and Cybernetics.
- Worked as a *Omdena* collaborator (ML Engineer) to solve the renewable energy AI challenge for African communities.
- Received a certificate of participation for the implementation of the theme 'Launch a Module' in the 2016 e-Yantra Robotics Competition 2016.
- Co-led the team to come up with a full-fledged product that can raise the standard of living in our society in six months. Our project, "Dexterous Service Robot," was one of the top seven finalists in the Anveshan Fellowship 2018, organized by Analog Devices, and my team did a real-time demo at the event.
- Presented a paper entitled "Colour based Object Sorting Robotic arm using Image Processing" in *National Conference on Robotics, Automation, Control and Embedded Systems (NCRACES-2017).*
- Led a team for the 2017 e-Yantra Robotics Competition and other robotics projects at the e-Yantra Robotics Lab at SJCE.
- Volunteered to teach children at the U&I charitable organization. Received the "Best Teacher of the Year 2019-20" recognition.
- Volunteered in the technical activities of the IEEE-SJCE student branch for the years 2015-2018.

LANGUAGES

- French (fluent, B1), English (fluent), Kannada (native), Hindi (beginner)